

JESS Thermodynamic Database v8.9

Mendelevium

24-Jan-23

Reaction No. 55639							
Md+3 + e-1 = Md+2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-0.1 (1SF)	0	CRV	6330
Note (1): For inter- and intra-reaction consistency (looks like a round-off error)							
2	25	0	Inf. Dilution	E:-0.15 (2SF)	3	ENS	7276
3	25	0	Inf. Dilution	E:-0.1 (1SF)	0	CRV	7761
4	25	0	Inf. Dilution	E:-0.15 (2SF)	3	CRV	22551
5	25	0	Unknown	E:0.6 (0.5SD)	0	ENS	5256
6	25	0	Inf. Dilution	dH:55.7 (3SF) kJ	3	CRV	7761

Reaction No. 71083							
Md+2 + 2<e-1> = Md(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.40 (3SF)	4	CRV	6330
2	25	0	Inf. Dilution	E:-2.40 (3SF)	3	CRV	7761
3	25	0	Inf. Dilution	dH:451.6 (4SF) kJ	3	CRV	7761

Reaction No. 71084							
Md+3 + 3<e-1> = Md(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-1.65 (3SF)	2	CRV	6330
2	25	0	Inf. Dilution	E:-1.65 (3SF)	2	CRV	7761
3	25	0	Inf. Dilution	E:-1.74 (3SF)	0	CRV	22551
4	25	0	Inf. Dilution	dH:510.4 (4SF) kJ	2	CRV	7761

Reaction No. 72046							
Md+3_OH-1(3)_(s) + e-1 = Md+2_OH-1(2)_(s) + OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-25.5 (3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.4 (2SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:109.2 (4SF) kJ	3	CRV	7761

Reaction No. 72047							
$\text{Md+3_OH-1(3)_(am.,s)} + 3\text{<e-1>} = \text{Md(s)} + 3\text{<OH-1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.13(3SF)	3	CRV	7761
2	25	0	Inf. Dilution	dH:540.(3SF)kJ	3	EES	
Note (2): By analogy with crystalline solid							

Reaction No. 72048							
$\text{Md+3_OH-1(3)_(s)} + 3\text{<e-1>} = \text{Md(s)} + 3\text{<OH-1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.17(3SF)	3	CRV	7761
2	25	0	Inf. Dilution	dH:543.3(4SF)kJ	3	CRV	7761

Reaction No. 72049							
$\text{Md+2_OH-1(2)_(s)} + 2\text{<e-1>} = \text{Md(s)} + 2\text{<OH-1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.5(2SF)	3	CRV	7761
2	25	0	Inf. Dilution	dH:424.9(4SF)kJ	3	CRV	7761

Reaction No. 79175							
$3\text{<OH-1>} + \text{Md+3} = \text{Md+3_OH-1(3)_(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.4(3SF)	1	ELC	

Reaction No. 79176							
$2\text{<OH-1>} + \text{Md+2} = \text{Md+2_OH-1(2)_(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.381(4SF)	1	ELC	

Reaction No. 79177							
$3\text{<OH-1>} + \text{Md+3} = \text{Md+3_OH-1(3)_(am.,s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.3(3SF)	1	ELC	

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CRV	Critical review
EES	Personal estimate
ELC	Linear Combination of Reactions
ENS	Not specified

References

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Advanced Inorganic Chemistry, 5th Edn., 1988 Cotton FA; Wilkinson G; John Wiley & Sons Inc., N.Y., U.S.A

Reference No.: 7761(89Bra)

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